Analog Electronics II: Op-amp & 555 Timer

Course instructor:
Dr. Mohsin Sajjad
Assistant Professor
North South University

EEE 311

Lecture # 6

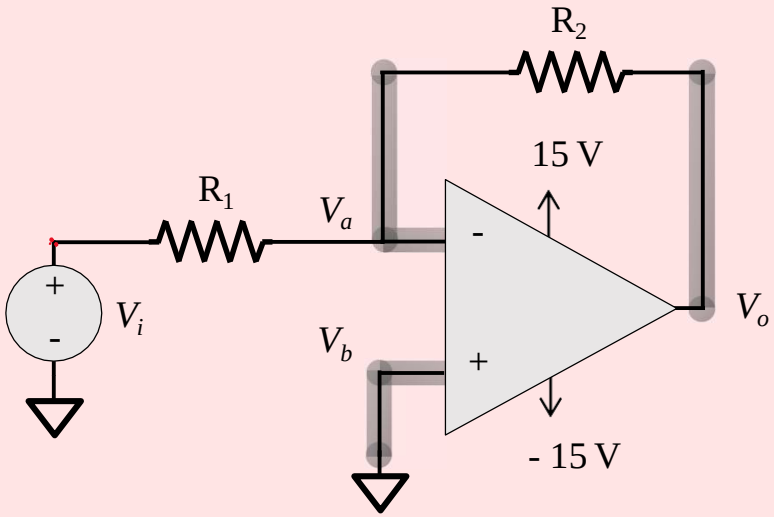
Inverting & non-inverting configurations

Inverting configuration

$$A_{CL} = -\frac{R_2}{R_1}$$

Input impedance = R_1

Output impedance = 0

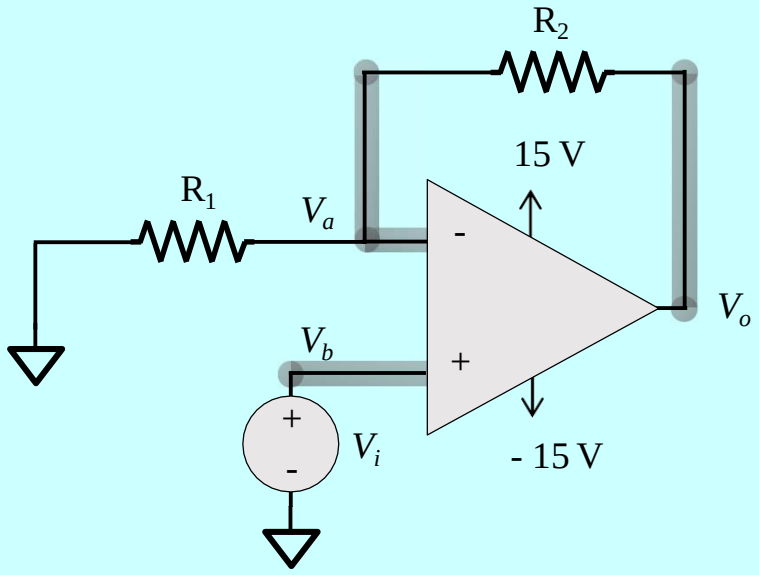


Non-inverting configuration

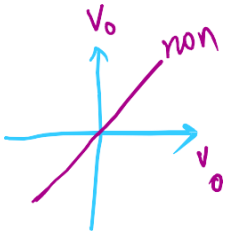
$$A_{CL} = 1 + \frac{R_2}{R_1}$$

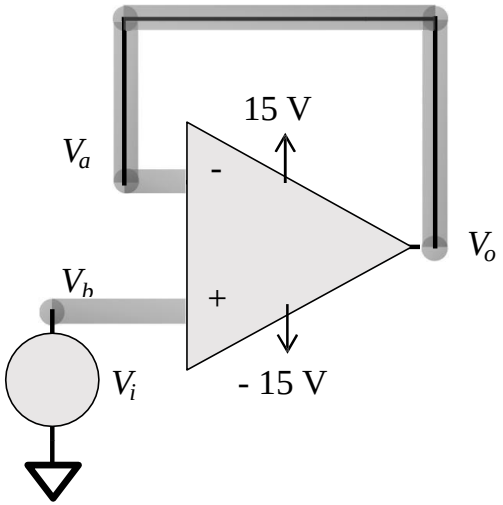
Input impedance = ∞

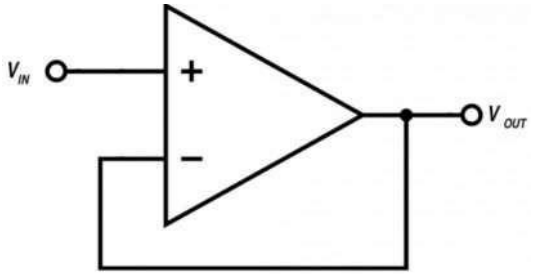
Output impedance = 0



Voltage follower – circuit diagrams



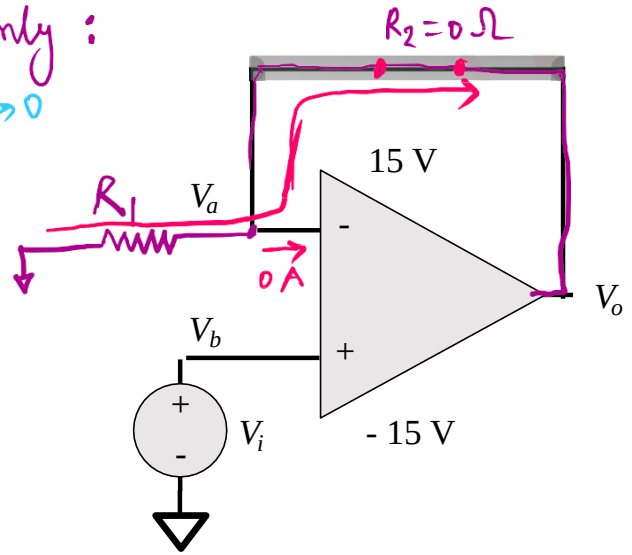




• $R_2 = 0\,\Omega$ only :

$A_{CL} = 1 + \frac{R_2}{R_1} \to 0$

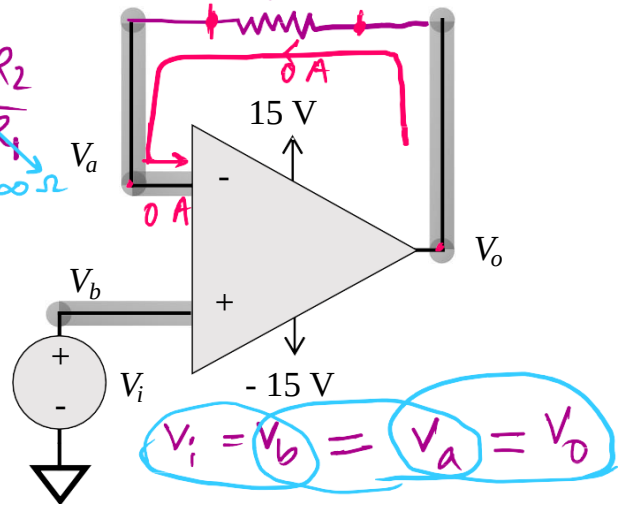
$= 1$



• $R_1 = \infty\,\Omega$ only : + 0 V −

$A_{CL} = 1 + \frac{R_2}{R_1} \to \infty\,\Omega$

$= 1$



ALL ABOUT CIRCUITS

Voltage Follower

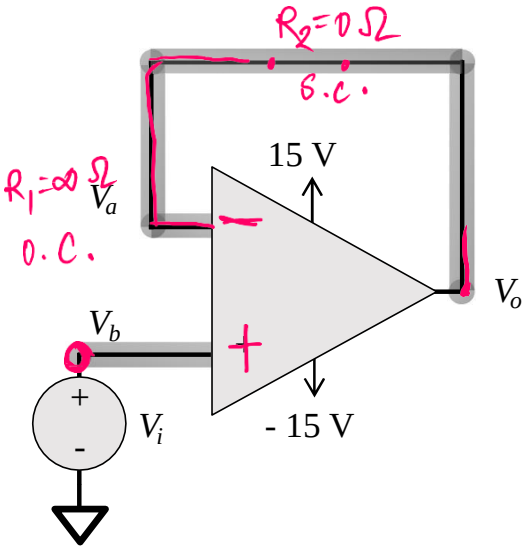
- -ve feedback
- Non-inverting configuration
- Unity gain

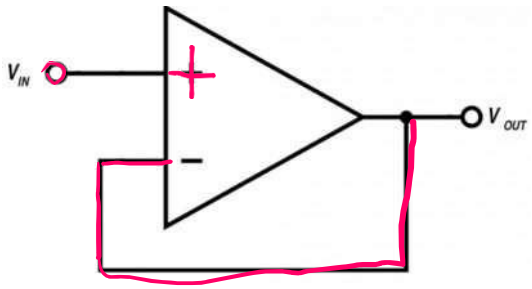
- Also called: **Unity-gain buffer** or **noninverting buffer.**
- The only necessary component is the op-amp itself.
- A voltage follower produces an output signal that is equal in amplitude to the input signal.
- Since the input signal is applied to the noninverting input terminal, no inversion takes place.

$A_{CL} = 1$ (keep)

$\frac{R_2}{R_1}$ • 0 Ω : S.C. only • Both (most common) • ∞ Ω : O.C. only

• both $R_2 = 0\,\Omega$ & $R_1 = \infty\,\Omega$



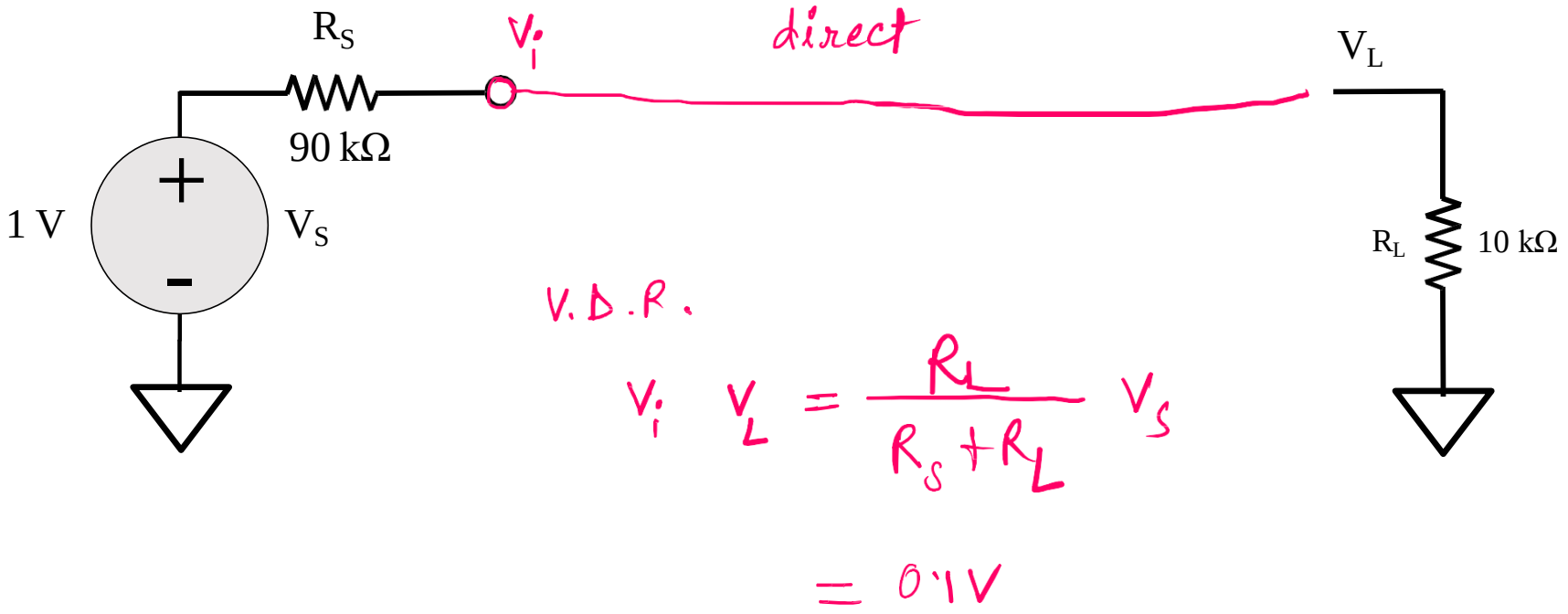


Source: ALL ABOUT CIRCUITS

Voltage follower -applications

- Effective voltage transfer requires a source circuit with low output impedance and a load circuit with high input impedance.

Model of a whole bunch ckt element



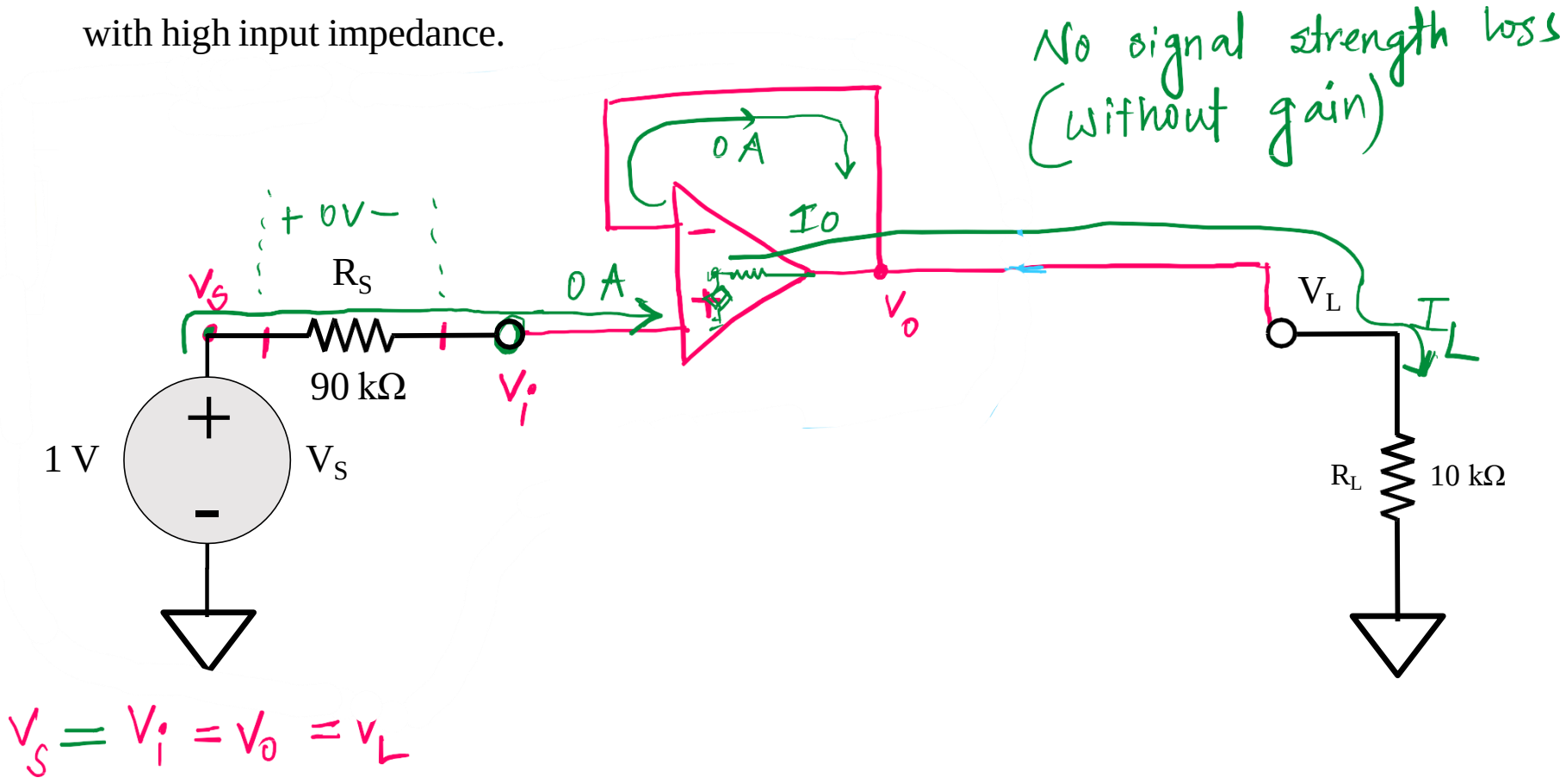
V.D.R.

$$V_i \quad V_L = \frac{R_L}{R_S + R_L} V_S$$

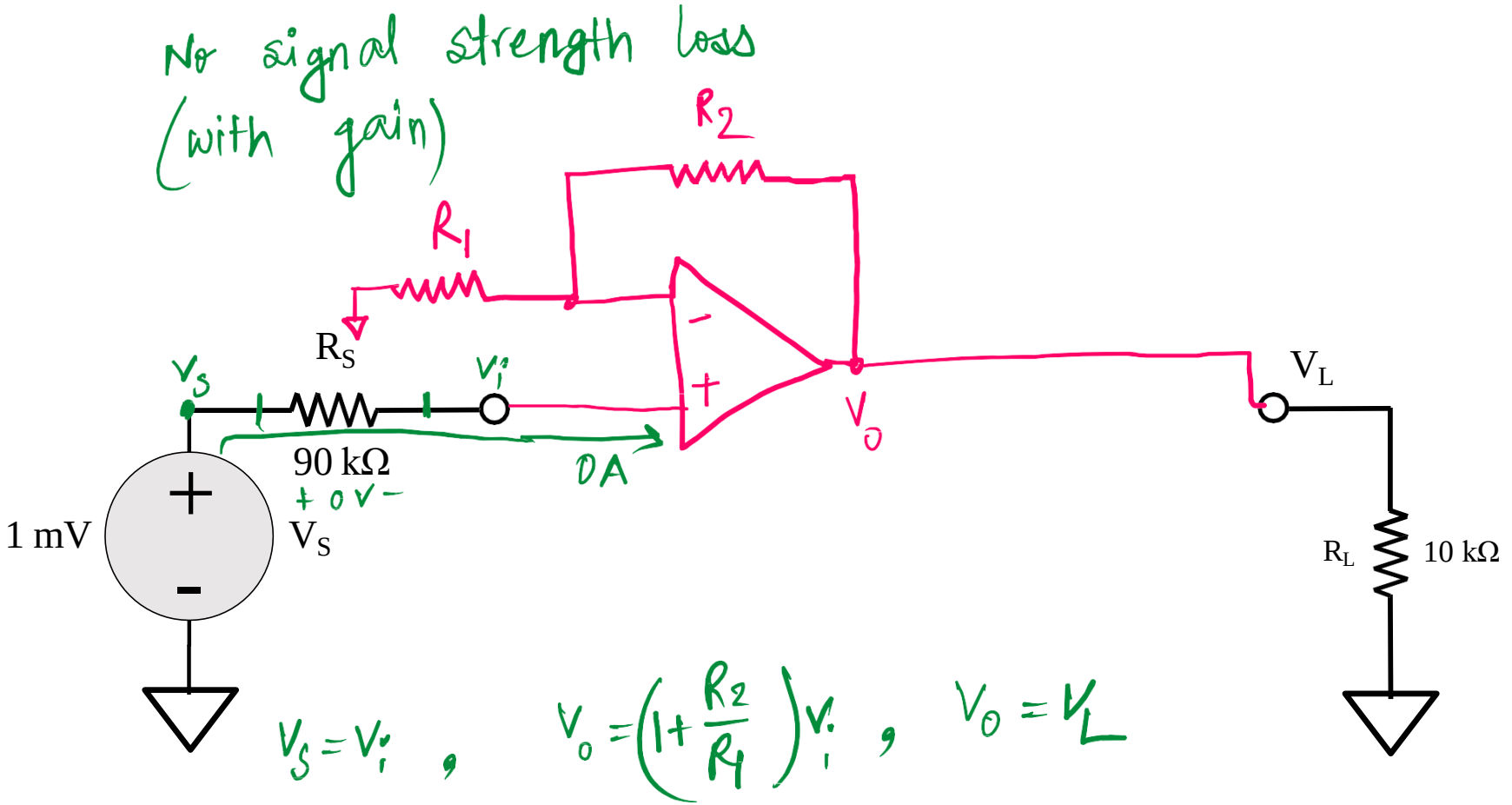
$$= 0.1V$$

Voltage follower -applications

- Effective voltage transfer requires a source circuit with low output impedance and a load circuit with high input impedance.

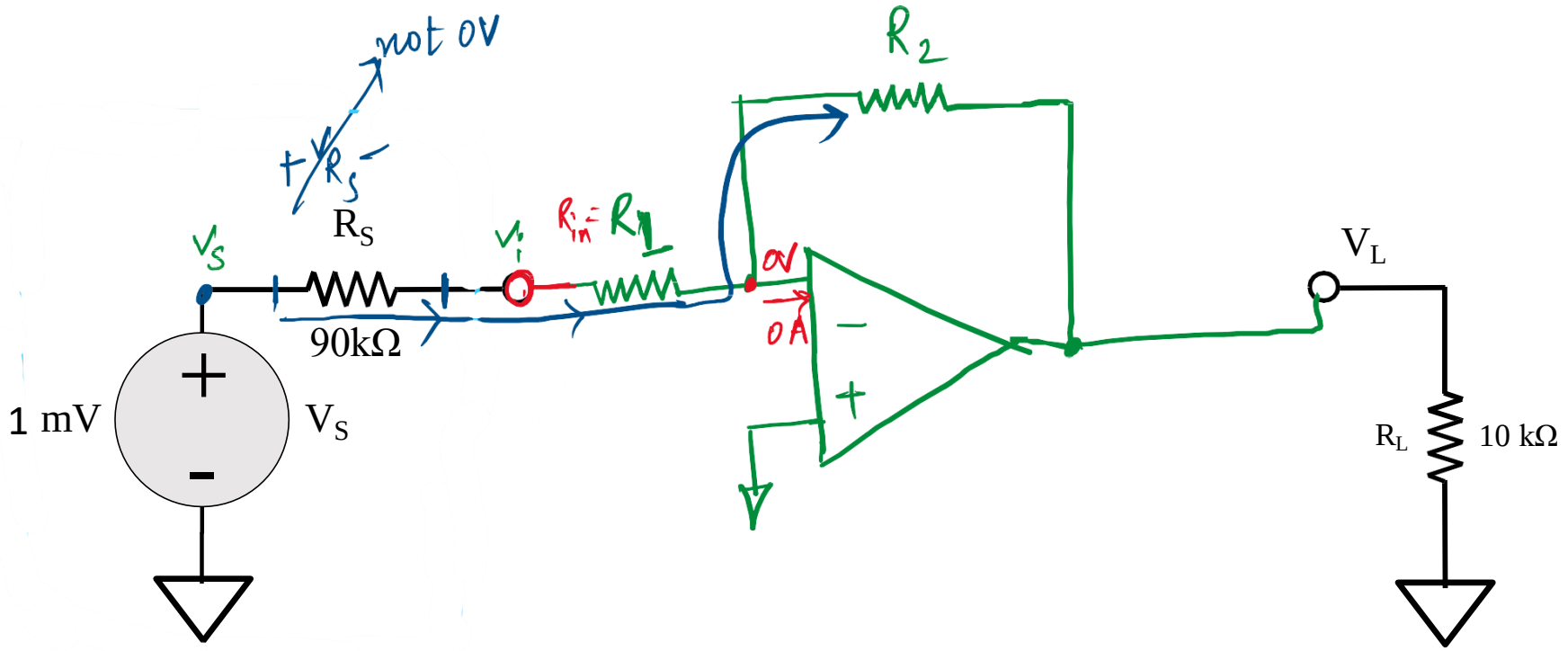


Non-inverting configuration - advantage



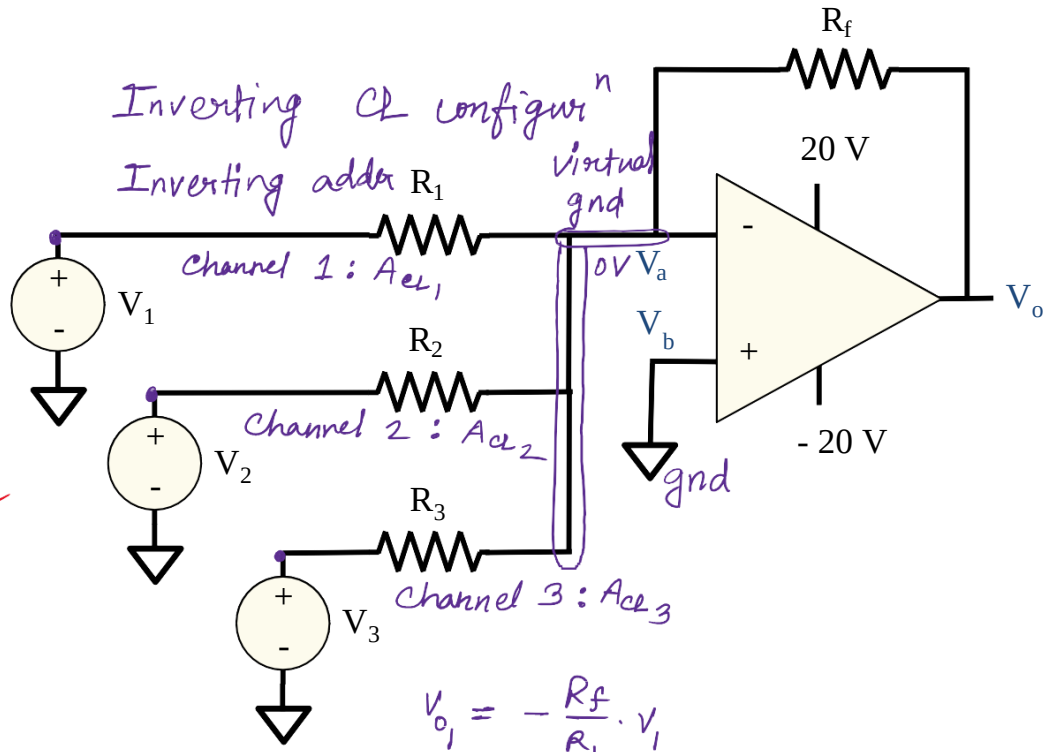
Inverting configuration - disadvantage

'inv. amp. ⇒ signal strength loss



Weighted Adder

Also called Multi-channel amplifier.



$$v_o = A_{CL_1} v_1 + A_{CL_2} \cdot v_2 + A_{CL_3} \cdot v_3$$

$$v_o = \left(-\frac{R_f}{R_1}\right) v_1 + \left(-\frac{R_f}{R_2}\right) v_2 + \left(-\frac{R_f}{R_3}\right) v_3$$

Memorize

$$A_{CL_1} = -\frac{R_f}{R_1}$$

$$A_{CL_2} = -\frac{R_f}{R_2}$$

$$A_{CL_3} = -\frac{R_f}{R_3}$$

$$V_{o_1} = -\frac{R_f}{R_1} \cdot V_1$$

$$V_{o_2} = -\frac{R_f}{R_2} \cdot V_2$$

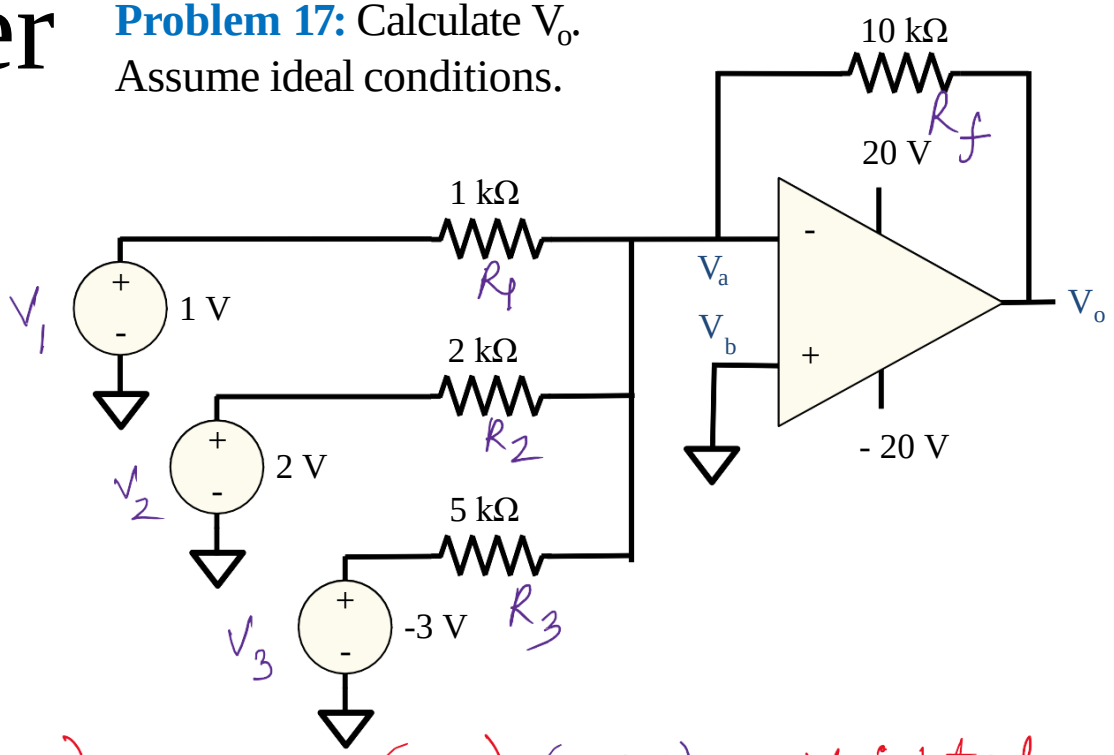
$$V_{o_3} = -\frac{R_f}{R_3} \cdot V_3$$

4 unknown variables, 3 eqn

assume 1 variable → most likely R_f

Weighted Adder

Problem 17: Calculate V_o. Assume ideal conditions.



$$v_o = \left(-\frac{10\,k\Omega}{1\,k\Omega}\right) \times 1\,V +$$

$$\left(-\frac{10\,k\Omega}{2\,k\Omega}\right) \times 2V +$$

$$\left(-\frac{10\,k\Omega}{5\,k\Omega}\right) \times (-3V)$$

$$v_o = (-10) \cdot 1V + (-5) \cdot 2V + (-2)(-3V)$$

Weighted Addition

$$= -14\,V$$

Source: Coughlin 3-2.1

Weighted Adder

Problem 18a: Design an amplifier so that the output v_o is related with the inputs v_1, v_2, v_3 in the following way:

$v_o = -(10v_1 + 5v_2 + 2v_3)$

Assume ideal conditions.

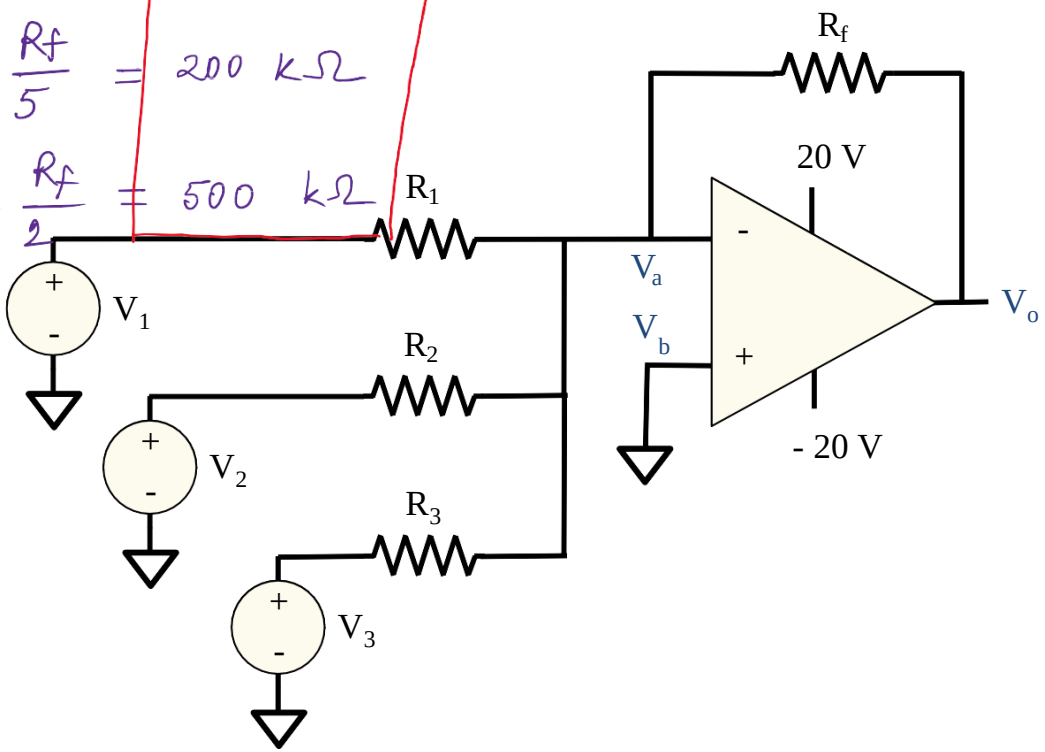
$R_f = 1\,M\Omega$ (assume)

$v_o = (-10)v_1 + (-5)v_2 + (-2)v_3$

$A_{CL_1} = -10 = -\frac{R_f}{R_1} \Rightarrow R_1 = \frac{R_f}{10} = 100\,k\Omega$

$A_{CL_2} = -5 = -\frac{R_f}{R_2} \Rightarrow R_2 = \frac{R_f}{5} = 200\,k\Omega$

$A_{CL_3} = -2 = -\frac{R_f}{R_3} \Rightarrow R_3 = \frac{R_f}{2} = 500\,k\Omega$



Design Steps

1. Identify the type of ckt from the given transfer eq^n.

$v_0 = -10v_1 - 7v_2$ — inverting adder

$v_0 = +10v_1 + 5v_2$ — non-inverting adder

$v_0 = -10v_{in}$ — inverting CL amp

$v_0 = +10v_{in}$ — noninv CL amp

$v_0 = +15V$ or $-15V$ — rectangular

saturating ckt : open-loop configur^n

2. Draw the ckt.
3. Label all resistors, voltages, currents that you are using in the calculation.
4. Write the relavant eq^n. Assume, one or more variables. Calculate rest of the variable.

Weighted Adder

Problem 18b: Design an amplifier so that the output v_o is related with the inputs v_1, v_2, v_3 in the following way:
$v_o = -(7v_1 + 5v_2 + 3v_3)$
Assume ideal conditions.

Use the method of the previous solution.

For the assumption of R_f in these problems,
you can take the largest multiple of LCM $\leq 10^6$

LCM: $7 \times 5 \times 3 = 105$

You can assume $R_f = 105 \times 9\ k\Omega = 945\ k\Omega$
to stay within range

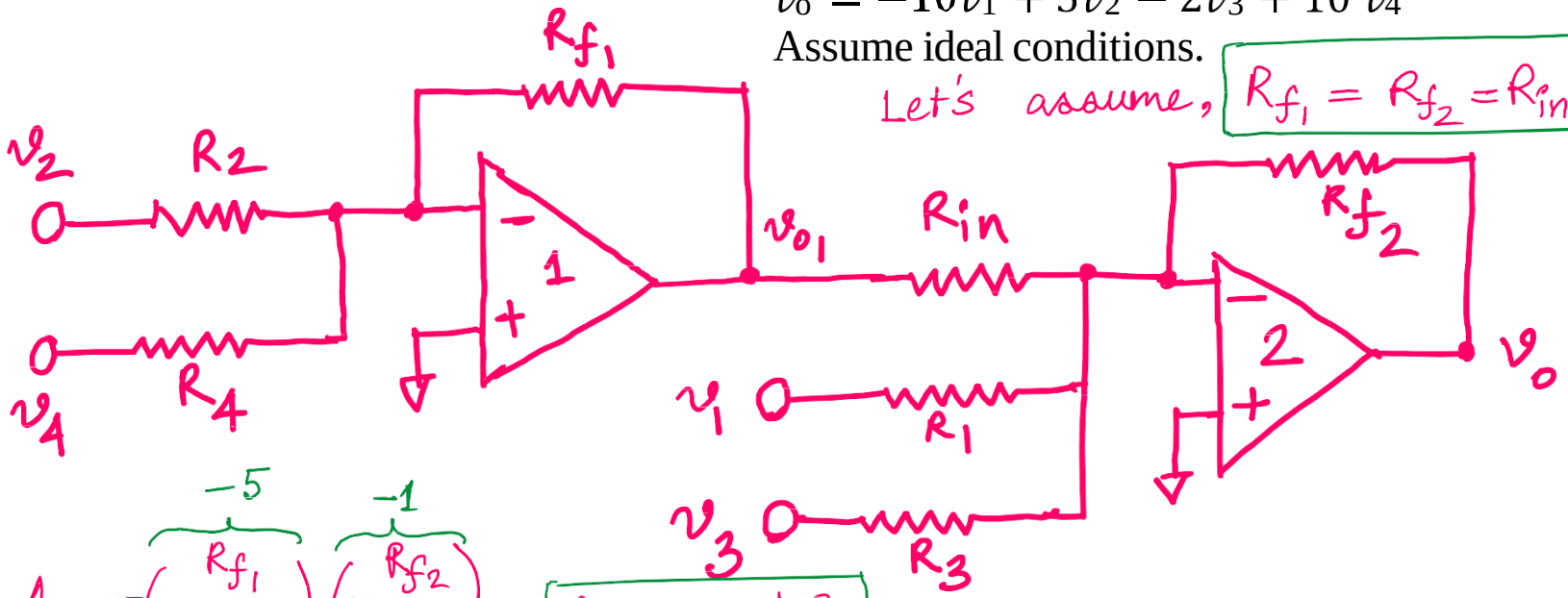
Weighted Adder

Problem 18c: Design an amplifier so that the output v_o is related with the inputs v_1, v_2, v_3 in the following way:

$v_o = -10v_1 + 5v_2 - 2v_3 + 10\ v_4$

Assume ideal conditions.

Let's assume, $R_{f_1} = R_{f_2} = R_{in} = 1\ M\Omega$



$A_{CL_2} = \left(-\frac{R_{f_1}}{R_2}\right)\left(-\frac{R_{f_2}}{R_{in}}\right)$ (-5, -1) $\Rightarrow$ $R_2 = 200\ k\Omega$

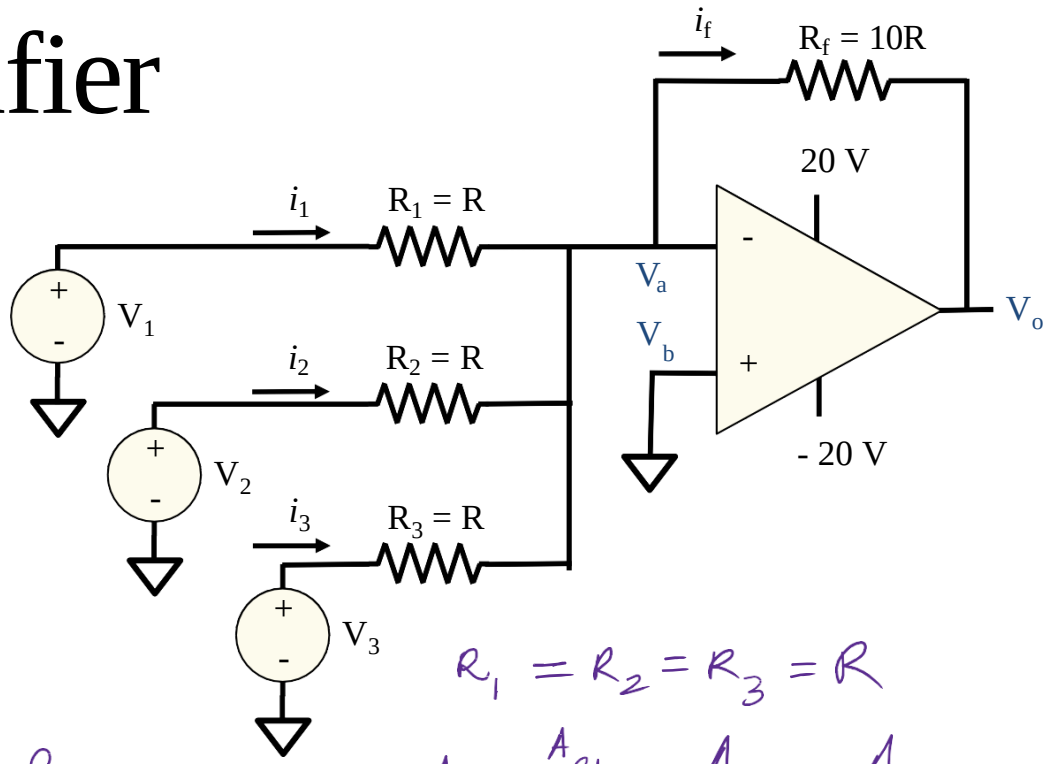
$A_{CL_4} = \left(-\frac{R_{f_1}}{R_4}\right)\left(-\frac{R_{f_2}}{R_{in}}\right)$ (-10, -1) $\Rightarrow$ $R_4 = 100\ k\Omega$

$A_{CL_1} = -\frac{R_{f_2}}{R_1} = -10 \Rightarrow R_1 = 100\ k\Omega$

$A_{CL_3} = -\frac{R_{f_2}}{R_3} = -2 \Rightarrow R_3 = 500\ k\Omega$

Summing Amplifier

Problem 19: Calculate V_o as a function of V_1, V_2, and V_3. Assume ideal conditions.



$$v_o = -\frac{R_f}{R_1}V_1 - \frac{R_f}{R_2}V_2 - \frac{R_f}{R_3}V_3$$

$$R_1 = R_2 = R_3 = R$$

$$A_{CL} = A_{CL_1} = A_{CL_2} = A_{CL_3}$$

$$v_o = -\frac{R_f}{R}V_1 - \frac{R_f}{R}V_2 - \frac{R_f}{R}V_3$$

$$= -\frac{R_f}{R}(V_1 + V_2 + V_3)$$

$$= \underbrace{A_{CL}}_{\text{amp}} \underbrace{(V_1 + V_2 + V_3)}_{\text{add/sum}} = -10(V_1 + V_2 + V_3)$$

Inverting Adder (direct addition)

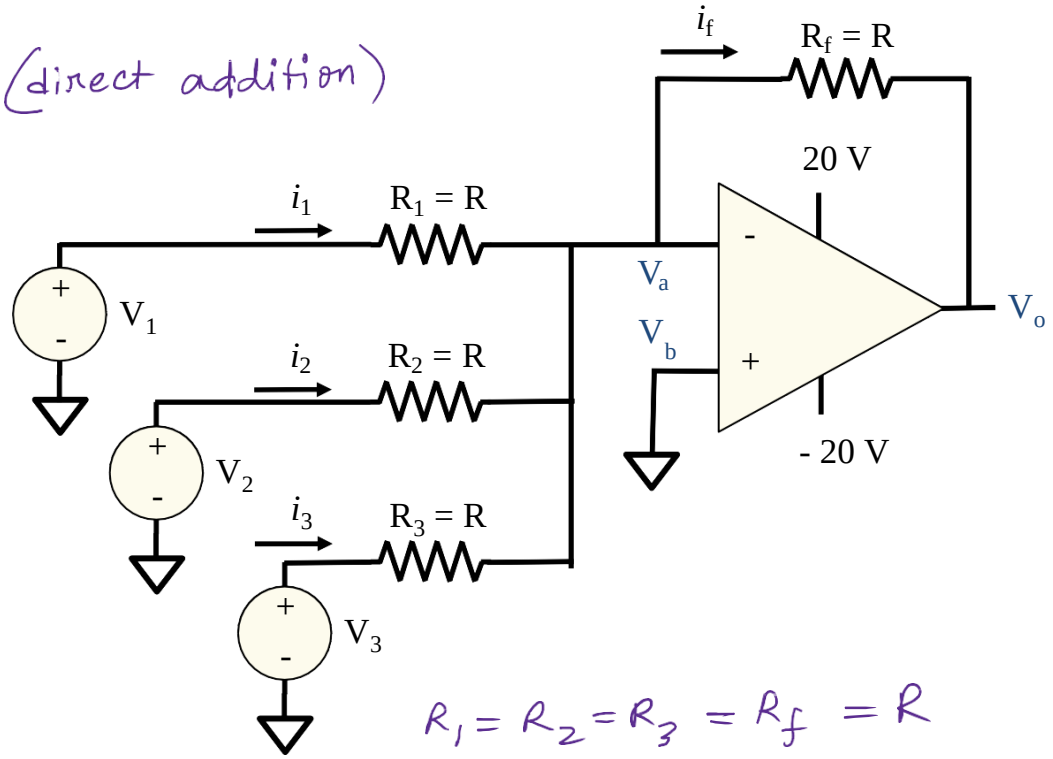
$$v_o = -\frac{R_f}{R_1} v_1 - \frac{R_f}{R_2} v_2 - \frac{R_f}{R_3} v_3$$

$$= -\frac{R}{R} v_1 - \frac{R}{R} v_2 - \frac{R}{R} v_3$$

$$= -v_1 - v_2 - v_3$$

$$= -(v_1 + v_2 + v_3)$$

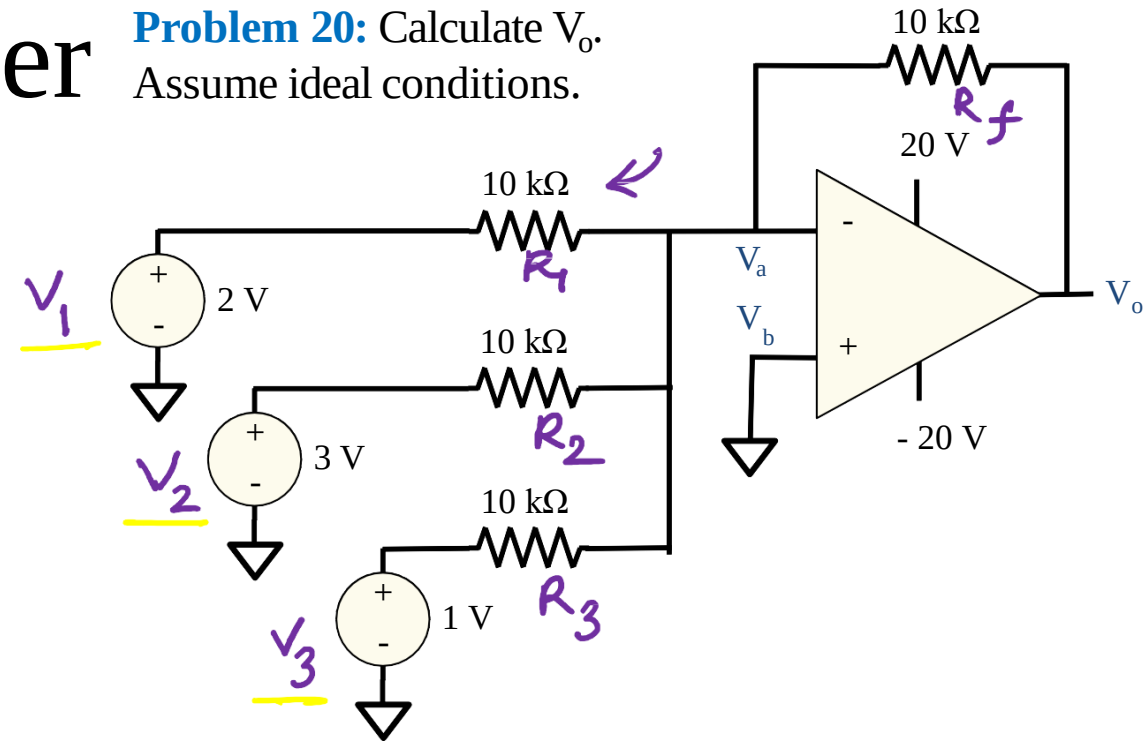
direct addition



$R_1 = R_2 = R_3 = R_f = R$

Inverting Adder

Problem 20: Calculate V_o. Assume ideal conditions.



$$V_o = -(V_1 + V_2 + V_3)$$

$$= -(2+3+1)\ V$$

$$= -6\ V$$